

03_SYSTEM_VALIDATION_AND_TEST_PLAN

Functional, decision, performance, failure and security validation plan.

Functional tests

Camera starts and recovers from disconnect. YOLO loads and produces detections. Tracking assigns persistent IDs. Polygon zone can be created and edited. Person outside zone does not trigger intrusion. Person inside zone triggers according to configured confidence/temporal rules.

Snapshot, prebuffer and post-event video are produced. Supabase event metadata is inserted and media is uploaded. Dashboard retrieves and displays the event.

ESP32 connects, reports heartbeat, receives buzzer command and recovers after disconnect. Missing radar/ground sensors can be simulated.

Decision tests

Person confidence 0.92 inside zone → immediate critical alert.

Person confidence 0.55 + radar ON + ground ON + temporal confirmation → confirmed event.

Person confidence 0.55 with radar/ground OFF → possible detection only.

Animal/bird/vehicle → informational event by default; no intrusion buzzer.

Radar without visual confirmation → suspicious-motion warning, never 'human detected'.

Ground trigger without visual confirmation → sensor warning, never 'human detected'.

Performance tests

Camera capture, YOLO inference, UI rendering, video encoding and cloud upload must be decoupled. Under overload, stale frames are dropped instead of building latency.

Measure FPS, inference latency, queue depth, dropped frames and upload latency. UI must remain responsive while uploads are running.

Failure tests

ESP32 offline: dashboard shows offline and local AI continues.

Supabase unavailable: local detection and evidence capture continue; upload is retried/queued.

YOLO model missing: clear startup error, no crash loop.

Camera unavailable: clear health error.

Storage upload fails: preserve local evidence and mark upload failure.

Security tests

No secrets committed. No service-role key in frontend. RLS remains enabled. Environment variables are used for credentials.

Final acceptance

Do not declare complete until the first demo sequence works end-to-end or every remaining failure is explicitly documented with logs and a next action.